

Environmental waste management and manufacturing of bioethanol for waste material of banana peels via fermentation

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Abstract. Worldwide, large quantities of fruit waste are generated by agricultural processes. Many times, this waste is thrown away in landfills. However, high sugars in fruit waste, such as sucrose, glucose and fructose, can be fermented to make bioethanol. Banana peels could be a great source for the generation of fuel bioethanol due to the vast amount of trash banana peels generate globally and their high lignocellulosic content. This study evaluated the effect of diluting H₂SO₄ at different concentrations (1%, 2% and 3%) with 2% cellulase in the hydrolysis process. It was found that the optimum condition for pineapple peel to produce a high amount of sugar was the sample treated under 2% H₂SO₄ hydrolyzed with 2% cellulase, which has total and fermentable sugar yields. Accordingly, this experimental study successfully produced the highest bioethanol after two days of fermentation. As a result of this, the study's findings revealed the right procedure for effectively converting the waste biomass from banana peels into bioethanol production.

Keywords: Waste management, Banana peels, Sugars, Fermentation, Bioethanol

1. Introduction

Increasing energy consumption, declining resources, increasing CO₂ emissions, and increased air pollution are all driving research and development of non-petroleum biomass. The use of biofuels to replace fossil-based transport fuels is a viable and sustainable option. Unlike fossil fuels which release CO₂ million years ago through combustion, the CO₂ released by biofuels is offset by the CO₂ captured in the growth of biomass, which has a much lower net effect on greenhouse gas emissions. In addition to the obvious benefits for the environment, the use of renewable raw materials as a replacement for fossil fuels can also have huge economic benefits. These include reducing dependence on crude oil, decreasing trade deficits, developing a strong biomass industry, and improving agricultural markets. The abundant resource of biofuels is biomass, which can be defined as industrial, agricultural, or forest resources (Manmai et al., 2019).

The high volume and low cost of fruit and agricultural scraps have led to a rise in the value of the high-value-added material conversion. Agroindustrial residue, including peels and seeds, makes up about half of all processed raw fruit. These fruit residues have little commercial value and are often disposed of at municipal dumps. However, they are an attractive choice for chemical production, as they provide a significant source of sugars. Because of their abundant and renewable nature, these materials are being used as raw materials to make competitive bioethanol for the open market (Ali et al., 2019). Therefore, bioethanol production from fruit residues may be an attractive option for disposing of these residues. Pectin, cellulose and hemicellulose are the main components of lignocellulosic feedstocks like fruit peels.

Because of their high carbohydrate contents, bagasse and orange peel waste could be used as feedstock for bioethanol production (Ramaraj et al., 2021). This is similar to other feedstocks.

Even though orange peel waste is important, there has been no research to examine the possibility of it being used in ethanol production. However, studies have shown other uses, including essential oil recovery, fermentation substrates for multienzyme preparations and pectin extraction. There has been an increase in waste treatment costs in the agroindustry (Cybulska et al., 2014). Although many studies have been done on the composition of banana pulp and how it can be used, most of the research on banana byproducts is focused on pectin extraction and phenolic compounds. Flavonoids and ethanol production are also being studied. Other studies have shown that banana pulp is high in cellulose, making it a promising source of polysaccharides. It also contains sugars like sucrose, glucose and fructose.

In order to convert banana peels into bioethanol or other lignocellulosic materials, a saccharification step must be first performed. This will convert the cellulose from the wastes into fermentable sugars like glucose. Different conditions can use with a variety of acidic and alkaline saccharification techniques. In order to prepare the banana peel for use in the manufacturing of bioethanol, a pretreatment process using acid-assisted ultrasonication was carried out. When producing bioethanol from lignocellulosic biomass, it is better to use biomass with a low lignin percentage and a high cellulose content. Enzymatic Saccharification is the most popular method of saccharification (Kuglarz et al., 2018; Manmai et al., 2020). Also, to make it easier for the enzyme to reach the cellulosic structures and remove interfering substances, a pretreatment step is typically performed prior to the enzymatic Saccharification step. After the saccharification step, fermentation is carried out to convert sugars and ethanol. Yeasts are used in this step. A better way is to use the enzyme for both saccharifications as well as fermentation. The enzyme converts cellulose into fermentable glucose, while the yeast performs the immediate conversion of sugars from ethanol. This method saves time, energy and costs, as well as reduces enzyme inhibition due to the lower immediate sugar concentration.

Important parameters in fermentation operations include the enzyme dosage, temperature, pH, and solid substrate loading. One of the problems to optimize this process is that the optimum temperature for the activity of the hydrolytic enzyme (about 50 °C for the cellulase) is usually different from the optimum temperature for the activity of the yeast, *Saccharomyces cerevisiae*. Many studies have investigated the effect of these parameters on the bioethanol production process. In the present study, therefore, banana peels waste biomasses with potential interest for bioethanol production.

2. Materials and Methods

2.1. Sample preparation

Samples of banana peels were taken from the fresh vegetable market in Chiang Mai, Thailand, near the Maejo University campus. The fruit peels were cut down to a smaller size to make it easier to use. They were left in the sun for two days. The substrate was then ground and placed in polyethylene bags. These bags were stored for further analysis. The samples were then analyzed for chemical composition and proximate analysis. This stage is allowed to determine the amounts of cellulose, water, hemicelluloses and lignin.

2.2. Pretreatment

The fruit wastes, i.e., fruit peels (20 g), were soaked in water for 2 hours and then autoclaved. The pretreated wastes were filtered and the filtrate obtained was analyzed for the total sugars and reducing sugars. The pretreated solid fruit waste substrate obtained from the above treatment was

dried and used for the subsequent hydrolysis studies using commercial cellulase for acidic hydrolysis. About 5 g of dried pretreated fruit wastes (peels) were used for the experiments in the present work.

2.3. Hydrolysis

Cellulase, a commercial enzyme, was used in this process. Hydrolysis was done at pH 4.5 at 50°C. The flask was kept in a closed container and given a general stir. The experimental setup was incubated for 72 hours with regular sample collection. After centrifuging the samples, the supernatant was taken and filtered sterilized. The hydrolysate was then filter sterilized, and the supernatant was collected in screw-cap bottles for storage.

The pretreated fruit peels (5g) were placed in conical flasks that contained dilute H₂SO₄ at different concentrations (0%, 1%, 2% and 3%, respectively). These were approximate twice the weight and volume of the fruits, i.e. 10 ml. The mixture was well mixed and allowed the peels to soak for two hours. These were then autoclaved to continue experiments. The liquid hydrolysates from the autoclave were then neutralised and recovered from the hydrolysis process. The hydrolysates samples were centrifuged. The solid fraction was washed with distilled water and dried. The supernatant was collected and used for the sugar determination.

2.3. Fermentation

The Vu et al. (2018) method was followed in order to carry out the fermentation process and determine the amount of ethanol produced. The robust yeast *Saccharomyces cerevisiae* was used in the process. Total and reducing sugar concentration estimated followed Vu et al. (2018) studies. The chemical and elemental composition data were presented using the mean and standard deviation (SD) of three replicates.

3. Results and discussion

Bioethanol is made from the distillation of the ethanol wash after fermentation of sugars derived from various biomasses. It can be used as a liquid fuel either in its pure form or in blends with petrol. Ethanol is high in octane, which allows it to achieve an octane-boosting or anti-knock effect similar to petroleum-derived aromatics such as benzene. Blend E85, 85% ethanol and 15% gasoline, has been used as an alternative fuel due to its advantages over pure ethanol (Bautista et al., 2018). Bioethanol is a new alternative fuel rapidly developing due to the rapid depletion of world fossil fuel reserves. There has been a lot of research into bioethanol's production and socioeconomic and environmental impacts.

Table 1. Chemical components of banana peels

Composition	Banana peel (wt%)
Cellulose	12.89±1.04
Hemicellulose	26.68±2.11
Lignin	7.99±1.05
Protein	6.23±2.27
Fibres	43.22±1.43

Large quantities of fruit waste are generated from the processing and selling of agricultural products (Chen et al., 2011). These include pear, banana, and apple residues after industrial processing. The most popular crop worldwide is bananas. Fruit peel waste is almost half of the total wet fruit mass after industrial processing. Annual banana production is million tonnes and 25-40% of this waste remains after processing (Choi et al., 2015). Although fruit waste is used as cattle feed due to its low protein content, it is not a valuable feedstock and is often disposed of in landfills or the ocean. These disposals can cause environmental problems because fruit waste is high in sugars and other nutrients. As a result, the cost of disposing of the waste is increasing. Fruit waste is high in sugars, including glucose, fructose and sucrose, as well as structural cellulose and hemicellulose (Khammee et al., 2021a,b). Compositional analysis of banana peels wastes is presented on a dry matter basis in Table 1. These chemical constituents and the abundance of fruit waste suggest that fruit is a good source for ethanol production.

Sugar yields after pretreatment and hydrolysis of banana peels are shown in Table 2. Because banana fruit peels and their accompanying residual biomass are both amylaceous and lignocellulosic components, they need to be hydrolyzed first in order to be transformed into glucose (Chuanchai & Ramaraj, 2018). Glucose can then be utilized as a feedstock for fermentation, which is the process by which ethanol is produced (Manmai et al., 2021a,b). Different types of banana waste were explored as possible sources of bioethanol production. For the production of bioalcohol from biomass, it is crucial to determine which conversion route to use (Sophanodorn et al., 2022a,b). All the possible conversion routes to bioalcohol from banana wastes involve destroying the lignin crystal structure and separating the cellulosic and hemicellulosic components. Enzymatic hydrolysis, which is biologically generated, ecologically benign, and highly efficient, hydrolyses cellulose and hemicellulose to produce fermentable sugars (Hsu et al., 2010). Cellulase is one of several cellulolytic enzymes used to hydrolyze cellulose. It also produces cellobiose and oligosaccharides. To complete the hydrolysis of cellulose, at least three enzymes must work together to break down the cellulose and give usable energy in the form of glucose (Wang et al., 2019).

Table 2. Sugar yields after pretreatment and hydrolysis of banana peels

Pretreatment	Loaded enzyme	Total sugar (g/L)	Reducing sugar (g/L)
0% H ₂ SO ₄	2% Cellulase	342.23±1.03	130.43±1.23
1% H ₂ SO ₄	2% Cellulase	365.44±2.04	147.27±1.03
2% H ₂ SO ₄	2% Cellulase	377.18±1.44	165.66±1.43
3% H ₂ SO ₄	2% Cellulase	370.54±0.78	157.53±1.29

The initial fermentability of these untreated hydrolysates by *S. cerevisiae* was extremely poor because of the presence of inhibitory agents. If this inhibition cannot be overcome by the selection or adaptation of resistant strain. Cellulases and hemicellulolytic enzymes possess particular sites for attachment to their substrate because this enzyme-substrate complex of one enzyme carries out slow degradation of another substrate (Wang et al., 2019). Therefore, preliminary experiments were performed to determine the optimum enzyme dose for maximum hydrolysis of banana peels. It will be economical, cost-effective as well as efficient for the saccharification of various polysaccharides of plant origin leading to the generation of fermentable sugars which can then be converted to many value-added products such as bioethanol (Nguyen et al., 2020; Nguyen et al., 2022). Fermentation is a microbial oxidation process that converts soluble

sugars into simple chemicals and produces alcohol as a by-product. Various bacteria and yeast metabolize monosaccharides (e.g., glucose and fructose) and disaccharides (e.g., maltose and sucrose), resulting in the generation of bioethanol from banana wastes. The efficiency of any microorganism as a bioethanol fermenter can be determined based on its proficiency under different process conditions such as osmotic tolerance, genetic stability, pH, temperature, feed concentration, residence time, nutritional needs, etc (Ramaraj & Unpaprom, 2019).

Table 3. Reducing sugar and ethanol concentration

Fermentation time (day)	Reducing sugar (g/L)	Ethanol (g/L)
0	165.66±1.43	00.00±0.00
1	110.40±2.49	42.20±2.10
2	91.20±1.21	68.63±1.05
3	87.22±2.44	55.34±1.22

As Table 3 demonstrated, the highest bioethanol production was observed at days of incubation having 68.63±1.05g/L of ethanol concentration. It was looked into as a feasible alternative to produce bioethanol from banana peels. It was demonstrated that the combination of dilute acid hydrolysis and ultrasonication pretreatment was extremely effective in eliminating lignin as well as releasing more cellulose and hemicellulose, all of which are essential for the production of bioethanol.

4. Conclusion

In the current research, the effects of applying different concentrations of H₂SO₄ to the pineapple fruit peels in the role of a pretreatment were analyzed. In addition, the efficiency of using three different enzymes in the hydrolysis process was evaluated and compared. Pretreatment of the sample with H₂SO₄ was found to be incompatible with the delignification of lignocelluloses, which led to the production of more fermentable sugars in the sample, as determined by the findings of the investigation. The findings of the investigation led the researchers to come to this conclusion. On the other hand, the addition of cellulase and *S. cerevisiae* led to an increase in the total quantity of sugars that were produced as a result of the hydrolysis process. In spite of this, the application of 2% cellulase in 2% H₂SO₄ pretreatment samples resulted in the production of the maximum quantity of sugars. The aspects of the fermentation process that are discussed in this article could be utilized for the production of bioethanol from banana peel on a large scale, which would result in a reduction in the overall cost of the process.

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